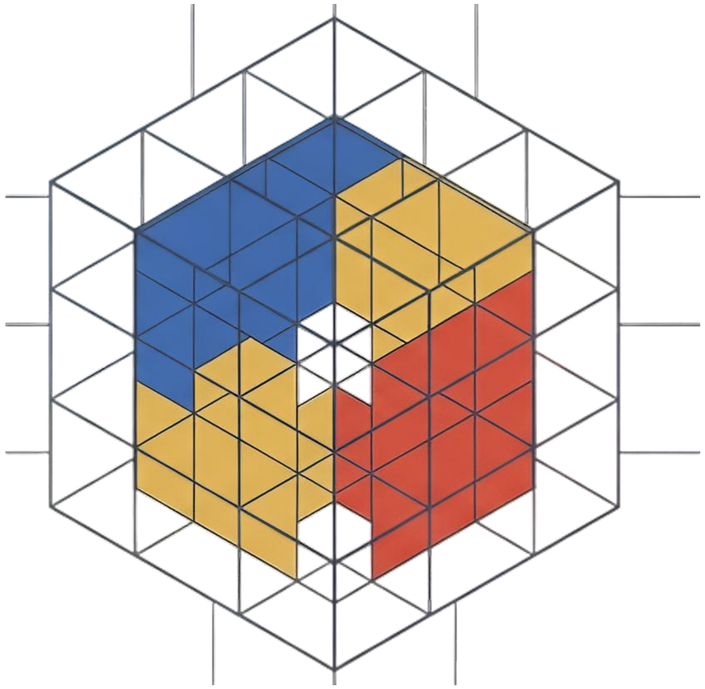
NOVA SCHOOL OF BUSINESS & ECONOMICS

# The Statistical Blueprint



A Conceptual Reference Guide to Variables, Parameters, and Probability Distributions.

Ana Amaro

Statistical Analysis

Accredited by

Member of

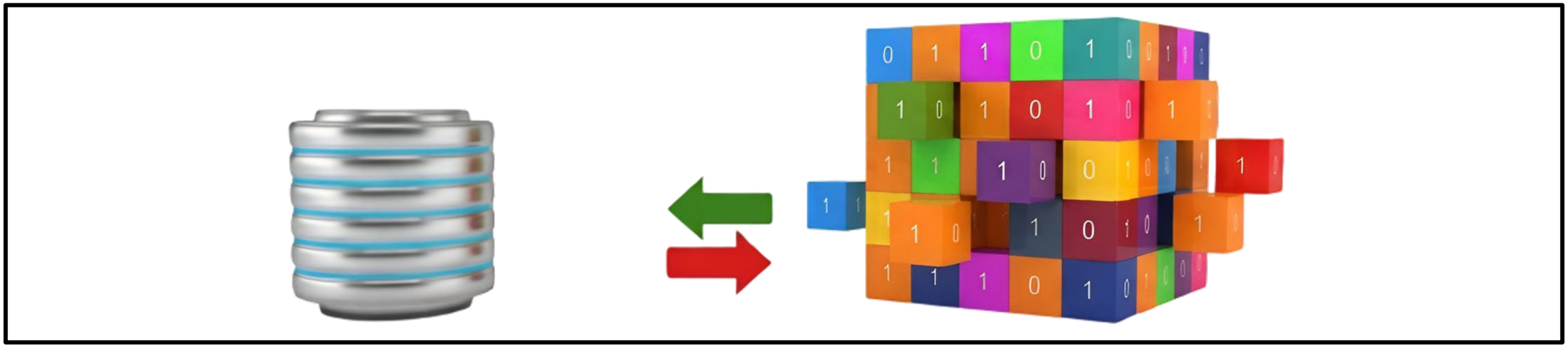
Participant in

Brand Recognition

Recovery and Resilience Plan (RRP)

Financiado pela União Europeia NextGenerationEU

# Data does not analyze itself.



Correct classification of variables guides appropriate statistical tools for summarizing, visualizing, and modeling data.

# QUANTITATIVE VARIABLES

Represent numerical values.

## DISCRETE

Isolated counts



## CONTINUOUS

Continuous measurements



# QUALITATIVE VARIABLES

Describe categories.

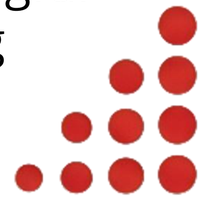
## NOMINAL

Categories without order



## ORDINAL

Meaningful ranking

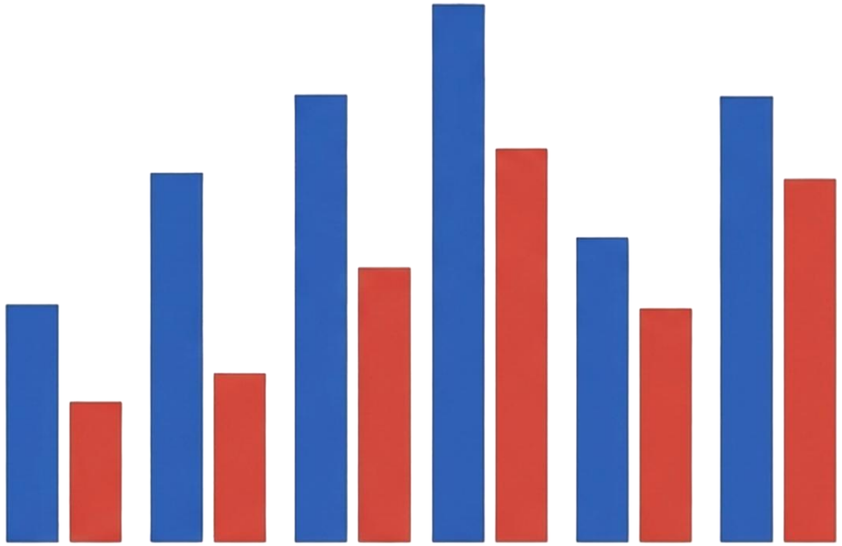


# Random Variables

Numerically represent outcomes of random processes to describe uncertainty mathematically.

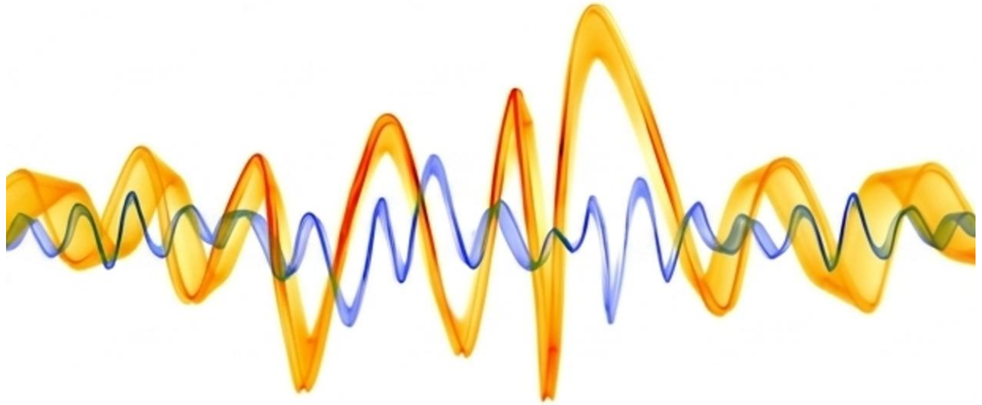
## Discrete Variables

Take isolated values with specific probabilities.



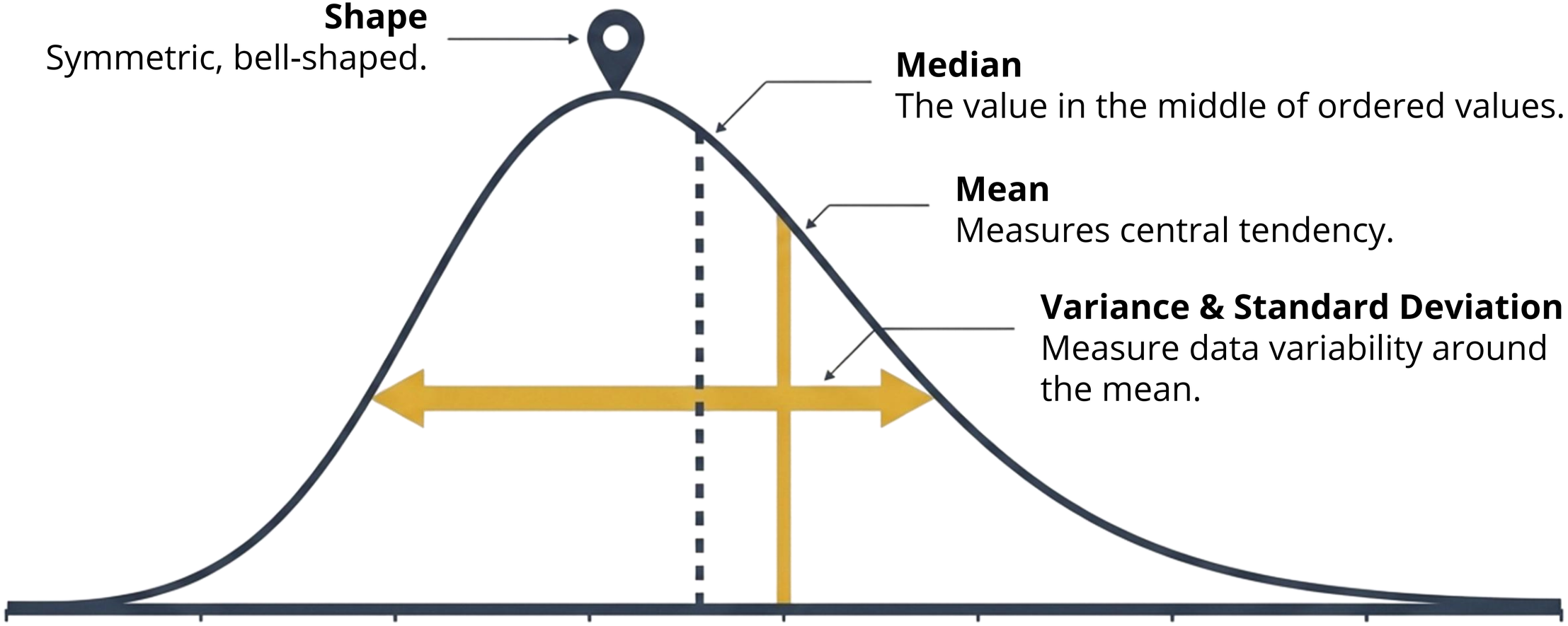
## Continuous Variables

Defined by probability density functions.



# The Parameter Anatomy

**Shape**
Symmetric, bell-shaped.



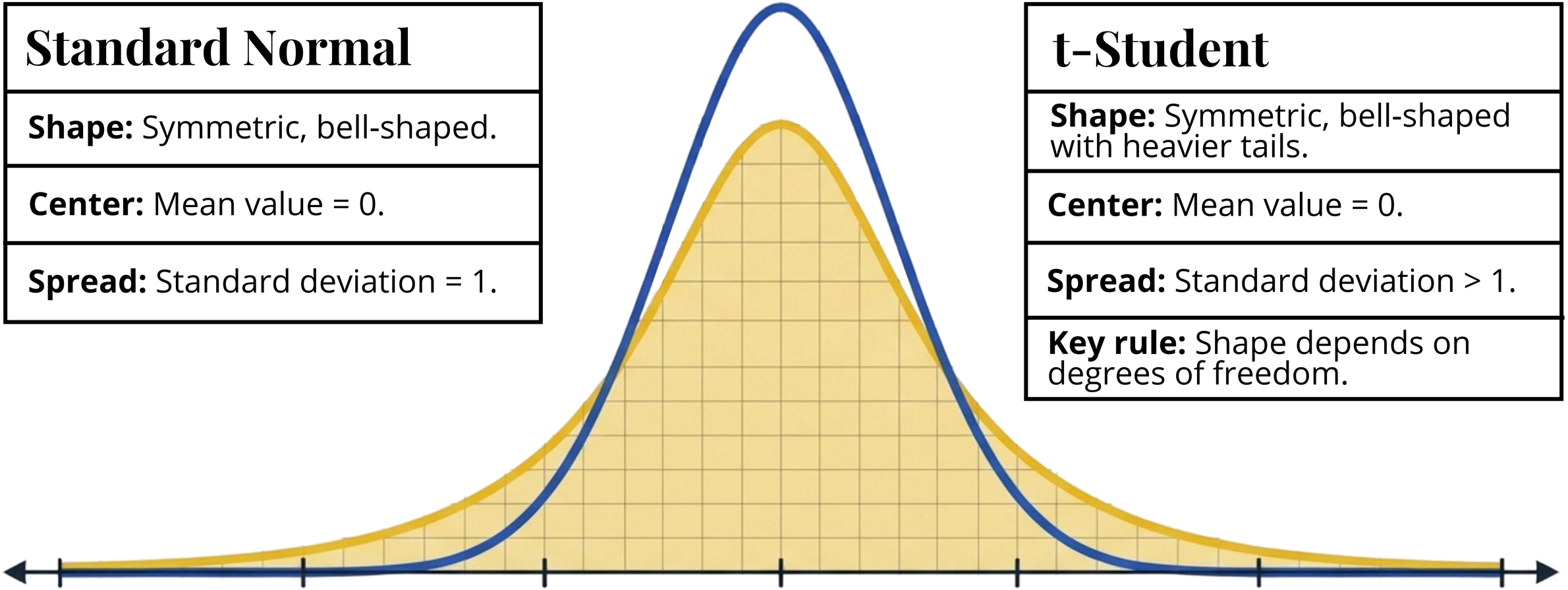
**Median**
The value in the middle of ordered values.

**Mean**
Measures central tendency.

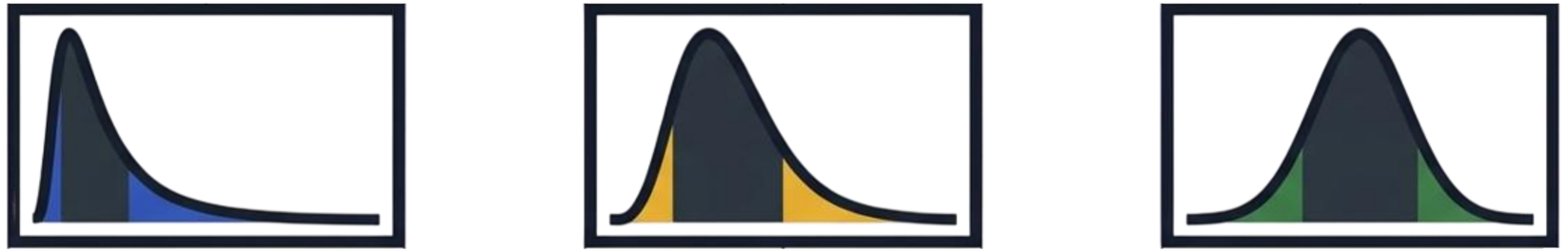
**Variance & Standard Deviation**
Measure data variability around the mean.

# **Probability distributions** assign probabilities to possible value of a random variable

| Standard Normal |
| --- |
| **Shape:** Symmetric, bell-shaped. |
| **Center:** Mean value = 0. |
| **Spread:** Standard deviation = 1. |



| t-Student |
| --- |
| **Shape:** Symmetric, bell-shaped with heavier tails. |
| **Center:** Mean value = 0. |
| **Spread:** Standard deviation > 1. |
| **Key rule:** Shape depends on degrees of freedom. |



As Degrees of Freedom increase...

## Chi-Square Distribution

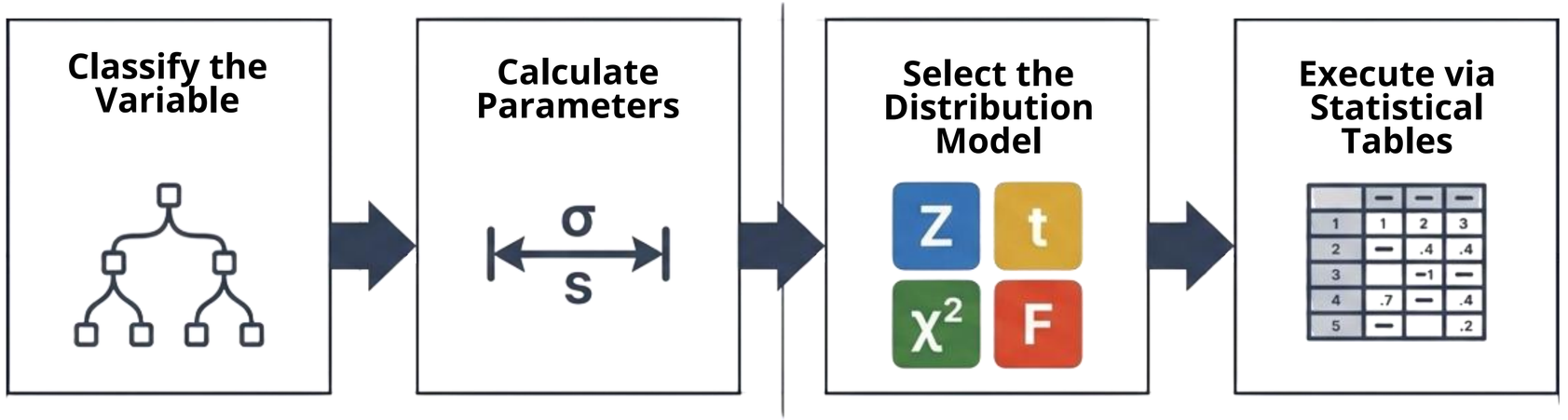
Not symmetric; inherently skewed to the right. Format depends directly on the number of degrees of freedom, becoming progressively more symmetric as they increase.

## F-Snedecor Distribution

Not symmetric; inherently skewed to the right. Format is highly complex as as it depends simultaneously on two distinct sets of degrees of freedom.

| | **Z**<br>Standard Normal | **t**<br>t-Student | **$\chi^2$**<br>Chi-Square | **F**<br>F-Snedecor |
|---|---|---|---|---|
| **Shape & Symmetry** | Symmetric Bell | Symmetric Bell (Heavy Tails) | Right-Skewed (Morphs) | Right-Skewed |
| **Mean** | 0 | 0 | N/A | N/A |
| **Spread/ Std Dev** | 1 | >1 | N/A | N/A |
| **Degrees of Freedom Impact** | None. Fixed universal shape. | Determines exact standard deviation and tail thickness | Determines primary shape; drives toward symmetry. | Driven by two distinct sets of degrees of freedom. |

# The Analytical Pipeline




**The end goal:** Utilizing expertise on Z, t-Student, Chi-square, and F-Snedecor tables to accurately find probabilities and scores based on your initial data classification.